

Design Report

Methodology and Design of a Paralleled-GaN Phase-Leg Building Block for Stacked Polyphase Bridge Traction Drives

75 V DC link · 100 A_{rms} · 2 × EPC2361 per switch · symmetric centre-driver layout

PCB-Design project, KTH

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Abstract

This report presents the design of a GaN phase-leg building block for stacked polyphase bridge (SPB) traction drives, structured after the design methodologies of recent paralleled-GaN inverter and power-module publications [1, 2, 3, 4, 5, 6]. The design is formulated as a multi-objective problem (efficiency, power density, switching frequency, voltage stress, current sharing, gate robustness) and solved in a model-based framework: analytical sizing, a parametric layout that drives Ansys Q3D extraction, LTspice switching simulation with the extracted *RLK* matrix, and loss, thermal and DC-link models that close the loop to the switching-frequency choice. The result is a 30 × 28 mm four-layer block with two mirrored commutation cells and a centre gate driver: 0.37 nH commutation loop per cell (Q3D, copper + MLCC ESL; package inductance not included), 1.70 nH matched gate loops, < 0.1 % predicted current imbalance, peak V_{DS} of 85.1 V at 160 A with $R_{g,on}/R_{g,off} = 2/0 \Omega$, and 99.58 % efficiency at 50 kHz (99.50 % at 100 kHz) for 2.65 kW per leg. The report also identifies where the design is close to its limits: the DC-link ripple, the high-side gate bump, and the loss with worst-case on-resistance.

Table 1: Design specification and key results.

Quantity	Value	Basis
DC link / phase current / rated power per leg	75 V / 100 A _{rms} (141 A _{pk}) / 2.65 kW	specification; $M = 1$, $\cos \varphi = 1$
Switching frequency	50 kHz nominal, 100 kHz capable	Sections 6, 12
Devices	2 × EPC2361 (100 V, 0.75/1.0 mΩ) per switch	Section 5
Commutation loop per cell	0.374 nH (incl. MLCC ESL, excl. package)	Q3D, Section 7
Gate loop per FET	1.70 nH, symmetric	Q3D, Section 8
Gate resistors on / off	2 / 0 Ω ($\zeta_{on} = 2.2$)	LTspice sweep + design window
Peak V_{DS} at 160 A; dv/dt on/off	83.1 / 85.1 V; 12 / 32 V/ns	LTspice with Q3D parasitics
Loss / efficiency at 50 kHz	11.12 W / 99.58 % (99.51 % max $R_{DS(on)}$)	Section 6
DC link per leg	24 × 10 μF 1210 + 12 × 1 μF 0805 (77 μF eff.)	Section 10
Board area, power density (board level)	8.5 cm ² , 0.31 kW/cm ²	Section 12

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1 Introduction

1.1 Motivation: why a low-voltage GaN building block for high-voltage traction

In a stacked polyphase bridge (SPB) the DC bus is divided into N series-connected modules, each supplying its own group of phases of a multiphase machine and seeing only V_{dc}/N . For an 800 V bus and 75 V modules ($N \approx 11$), 100 V GaN HEMTs can be used: they have the lowest $R_{DS(on)} \cdot Q_G$ figure of merit, no reverse recovery, and chip-scale packages with sub-nH package inductance. The low voltage step per module also reduces the winding insulation stress compared with a full-bus SiC inverter. Three-level GaN inverters [9] pursue a similar goal with 650 V devices. Because one building block is repeated N times, its efficiency, density and robustness multiply at system level.

1.2 Design challenges identified in the literature

- **Current capability** of a single GaN die is limited, so devices must be paralleled [2, 3, 4, 1].
- **Voltage overshoot** is the primary failure mechanism of low-voltage GaN modules. A commercial 100 V, 270 A-per-phase module failed at 292 A, 36 % of its die rating, because of its package inductance [6].
- **Asymmetric loops** give unequal dynamic current sharing. Lateral loops and cascaded phases give 11–28 nH unequal loops [6]. Mismatched quasi-common-source inductance and power-to-gate coupling are the most critical factors [1, 5].
- **Fragile gates**: $V_{GS,max}$ is +6 V and V_{th} is about 1 V, so gate ringing and Miller bumps must be controlled [1, 5, 7].
- **Efficiency and density** trade off through the switching frequency, the DC link and the heat sink, so they must be co-optimised rather than optimised separately [2].

1.3 Approach and contributions of this design

1. A model-based design framework (Fig. 2) in which a single parametric geometry drives the layout drawings, the Q3D extraction and the SPICE netlist. Every number in this report can be regenerated.
2. A symmetric “butterfly” building block in the spirit of [4]: two mirrored vertical commutation cells, a centre gate driver with equal gate stars and Kelvin returns, and a distributed MLCC bank.
3. Quantified design rules: a gate-resistor design window, an overshoot design space, a current-sharing sensitivity ranking, and DC-link and thermal bounds versus f_{sw} .
4. A board-level efficiency–density– f_{sw} analysis following [2], and a winding-ripple requirement that can be handed to the machine design.

2 State of the art

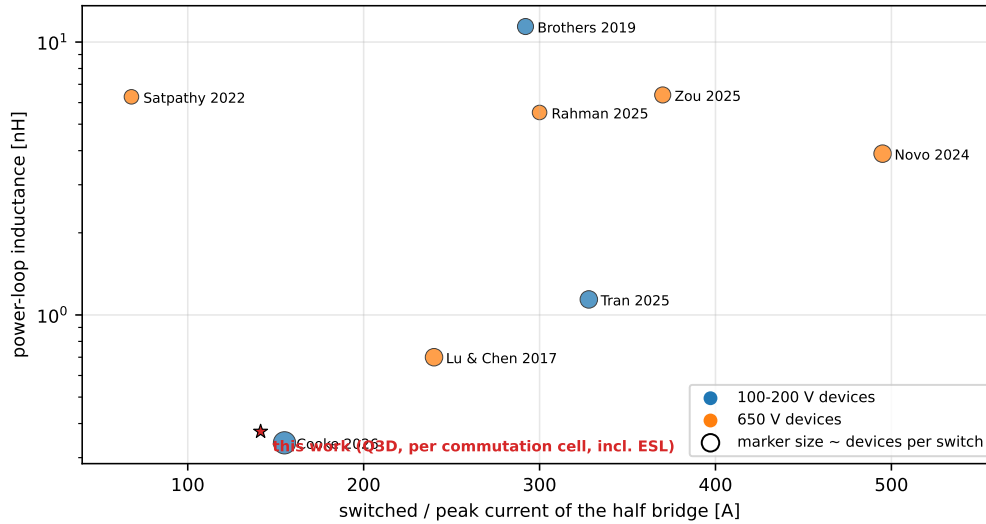
Table 2 and Fig. 1 summarise the reviewed paralleled-GaN half bridges. Two layout families stand out. PCB-based vertical (flux-cancelling) loops reach sub-nH inductance [5, 3]. DBC, IMS and module solutions trade inductance for thermal performance: 1.1–6.4 nH [1, 2, 11, 10]. The device power utilisation (DPU) of [3],

$$DPU = \frac{V_{dc} I_{ph}}{n V_{DS}^2 / R_{DS(on)}} \cdot 10^4, \quad (1)$$

is 3.75 for this design ($R_{DS(on),max}$; 2.81 with the typical value). This is comparable to the continuously operated inverters in [3] (2.3–9.5). Note that our value is a design point, not a test result.

Table 2: Reviewed paralleled-GaN half-bridge designs (values as reported; loop definitions differ between papers).

Work	Devices per switch	Test	L_{loop}	Gate drive	Key design features
Lu & Chen 2017 [5]	4 × GS66516T (650 V)	DPT 400 V/240 A	0.7 nH	one driver, symmetric	6-layer PCB, quasi-CSI <0.2 nH, analytic switching model
Brothers 2019 [6]	4 × GS61008 (100 V)	DPT 48 V/367 A	–	10/4 Ω per die	flux-cancelling PCB; commercial module failed at 292 A
Wattenberg 2023 [4]	8 × 100 V 3 mΩ	3-ph 48 V, 125 A _{rms}	–	centre driver, R_G 1 Ω	“butterfly” layout, gate loops ≈1 nH within 17 %, MLCC DC link
Musumeci 2022 [7]	2 × EPC2065 (80 V)	motor 48 V/35 A	–	10 Ω	V_{th} spread identified as the most critical parameter
Palma 2024 [8]	4 × EPC2302 (100 V)	motor 60 V/150 A _{rms}	–	one driver/leg	12-device characterisation board, peak-current prediction
Satpathy 2022 [9]	2 × GS66516B (650 V)	3L-ANPC 550 V, 70 kHz	6.3 nH	split $R_{\text{on}}/R_{\text{off}}$	IMS + capacitor PCB, PLECS thermal
Novo 2024 [10]	4 × D3GaN (650 V)	WLTP 350 A _{rms}	3.9 nH	drivers above devices	turn-on skew 2.8 ns, 99.3 %
Tran 2025 [1]	4 × GS61008P (100 V)	DPT 48 V/328 A, 5 kW	1.14 nH	Si8271, 4.7/4 Ω	IMS hybrid, Q3D→LTspice→Icepak framework
Zou 2025 [2]	3 × V08TC65 (650 V)	100 kW inverter	6.4 nH	integrated, 1.4 nH	η - ρ - f_{sw} Pareto co-optimisation
Rahman 2025 [11]	2 × GS-065-150 (650 V)	DPT 350 V	5.52 nH	co-packaged	multilayer substrate, 40 % lower L
Cooke 2026 [3]	8 × EPC2304 (200 V)	120 V/155 A, 100 kHz	0.34 nH (est.)	15/2 Ω, +5.3/−3 V	symmetric gate PCB, thermal-image sharing check
This design	2 × EPC2361 (100 V)	design 75 V/141 A	0.37 nH/cell	LT8418 centre, 2/0 Ω	mirrored cells, Q3D→LTspice, η - ρ - f_{sw} incl. machine

**Figure 1:** Power-loop inductance versus switched current for the reviewed designs. Low-voltage, high-current designs must sit at the lower right because $\Delta V = L di/dt$ has to fit into a 10–25 V margin.

3 Design objectives, constraints and methodology

3.1 Objectives and constraints

Table 3: Objectives and hard constraints.

	Metric	Target / limit
Efficiency	loss per leg at the rated point	$\leq 13.3 \text{ W}$ ($\eta \geq 99.5 \%$)
Switching frequency	highest f_{sw} meeting the efficiency target	$\geq 50 \text{ kHz}$; 100 kHz desirable
Winding ripple	$\Delta i_{\text{pp}} \approx V_{\text{dc}} / (4L f_{\text{sw}})$	requirement to the machine design (Section 6)
Power density	board area per kW (board level)	maximise
Voltage stress	peak V_{DS} (rating 100 V, 120 V repetitive transient)	$\leq 90 \text{ V}$ at 160 A (13 % overload)
Gate robustness	V_{GS} overshoot, Miller bump, negative excursion	$< 6 \text{ V}$; bump $\leq 1 \text{ V}$; $> -4 \text{ V}$
Current sharing	dynamic and static imbalance	as small as the layout allows
Measurability	DPT access to V_{DS} , V_{GS} and device current	test points without adding loop inductance

3.2 Trade-off matrix

Table 4 shows how each design variable acts on the objectives. Green means favourable, red unfavourable, yellow weak or mixed. It is the qualitative map that the following sections quantify.

Table 4: Effect of increasing each design variable on the objectives.

Increase of ...	Inverter loss	Winding ripple	Density	V_{DS} peak	Sharing risk	Gate bump	Cost
f_{sw}	↑	↓	↑ (DC link)	—	—	—	—
devices in parallel n	↓ cond., ↑ Q_{oss}	—	↓	↑ (longer loop)	↑	—	↑
L1–L2 dielectric h	—	—	—	↑	—	—	↓
$R_{\text{g,on}}$	↑ E_{on}	—	—	↓ (HS)	↓	↓	—
$R_{\text{g,off}}$	small	—	—	small	—	↑	—
copper weight (oz)	↓	—	—	slight ↑	—	—	↑
DC-link capacitance	ESR ↓	—	↓	↓ (ringing)	—	—	↑
gate-loop asymmetry	↑	—	—	—	↑↑	↑	—

3.3 Model-based design framework

Following the digital framework of [1] and the co-optimisation flow of [2], the design is carried out in the loop shown in Fig. 2. Unlike a layout-first flow, the geometry is a parametric Python description (`bb_geometry.py`). That one description generates the layout drawings, the Q3D model and therefore the SPICE parasitics, so a change of, for example, the dielectric thickness propagates to losses and overshoot without manual steps.

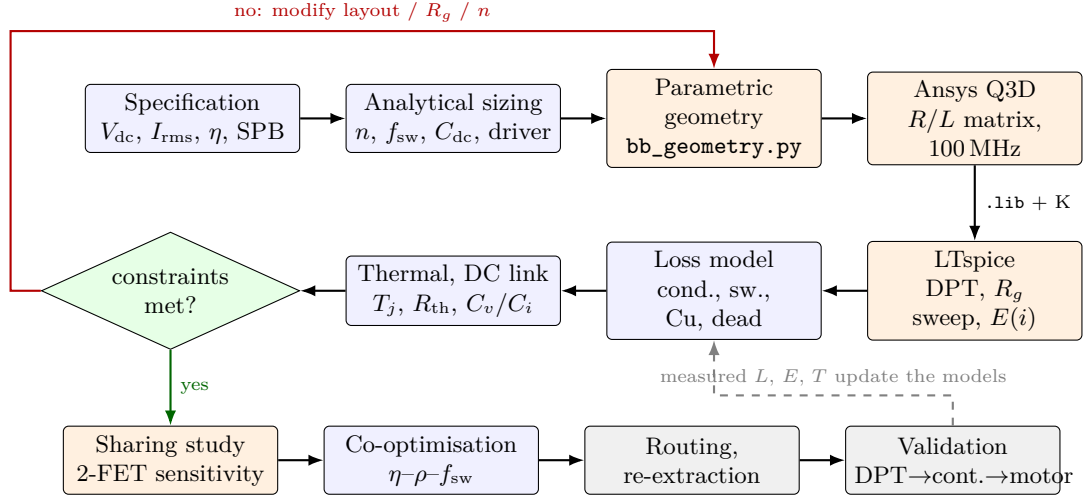


Figure 2: Design framework. Orange: field and circuit solvers; blue: analytical models; grey: hardware steps (pending).

4 System architecture

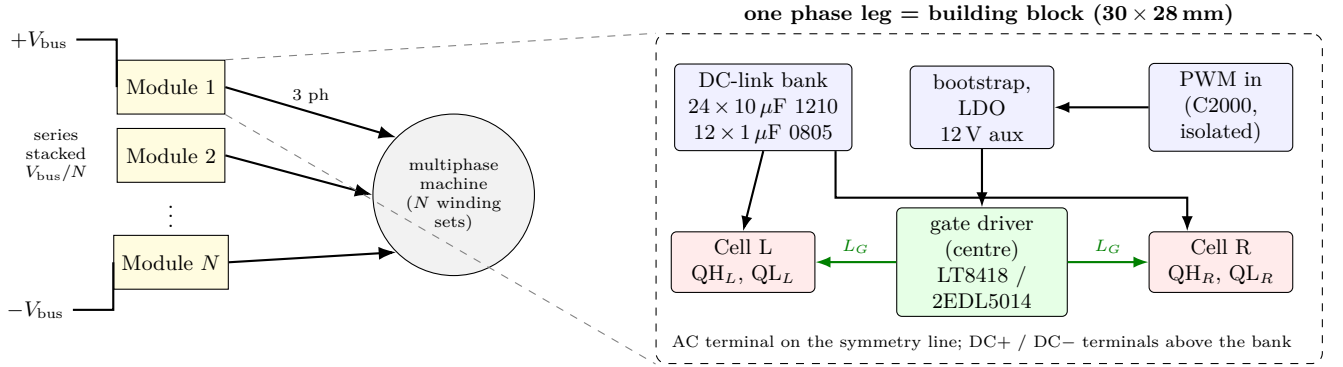


Figure 3: SPB system (left) and the building block (right). Each module holds three building blocks on a shared DC link; the block itself is mirror-symmetric about the driver corridor.

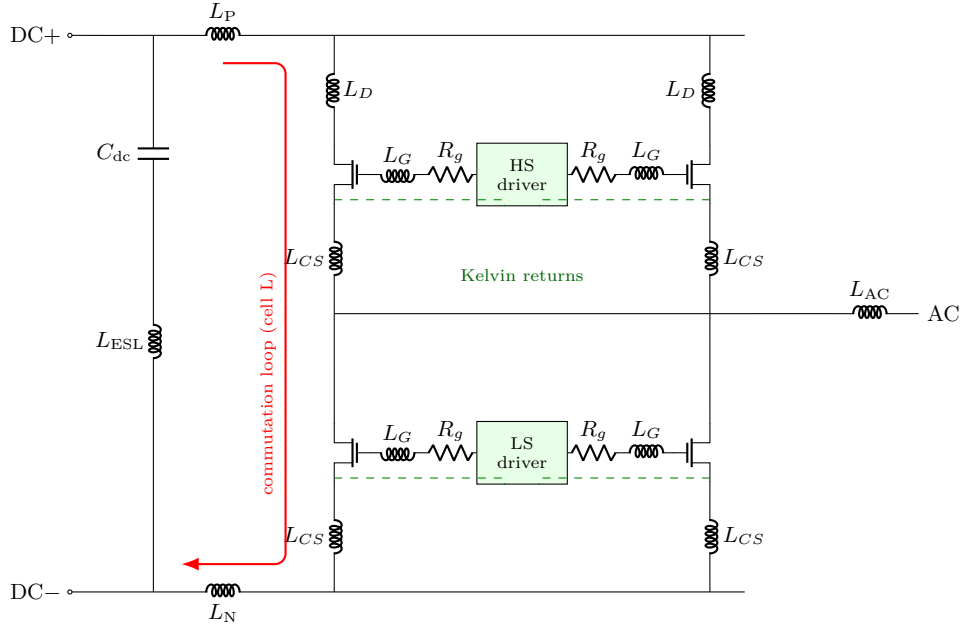


Figure 4: Equivalent circuit of the building block with the parasitic elements extracted by Q3D. The mirrored layout makes every left element equal to its right counterpart. L_{CS} is only the package and pad part because the gate loops return through Kelvin paths.

5 Device selection and paralleling

5.1 Device properties relevant to paralleling

- $R_{DS(on)}$ has a positive temperature coefficient ($\times 1.48$ at 100°C). This makes static sharing self-balancing [4, 5].
- V_{th} is 0.8 V min, 1.1 V typ and 2.5 V max (datasheet). The spread, not the temperature drift, sets the dynamic imbalance; it is the most critical parameter [7, 8].
- **Low C_{rss} (13 pF) and Q_{GD} (3.8 nC)** give fast Miller transitions, but also a high dv/dt coupling into neighbouring gates.
- $Q_{rr} = 0$ removes the dominant turn-on loss of Si. The Q_{oss} of 113 nC at 75 V remains and scales with n [5].
- **Top-side cooling** ($R_{\theta JC} = 0.2 \text{ K/W}$, source-connected lid) requires an insulating TIM, as in [4, 3].

5.2 Number of devices in parallel

A first bound follows [2]: $n \geq I_m k_T k_I / I_D$. With $I_m = 141 \text{ A}$, $I_D = 133 \text{ A}$, an assumed temperature derating $k_T \approx 1.25$ and the margin $k_I = 1.2$ of [2], this gives $n \geq 1.6$, i.e. $n = 2$. The loss-optimal n balances conduction ($\propto 1/n$) against the current-independent switching energy ($\propto n$). Fig. 5 shows that $n = 2$ is the knee. One device misses the target (16.2 W, 6.4 W per device). Three or four save only 1.3–1.7 W at the cost of a longer loop and a higher sharing risk [5].

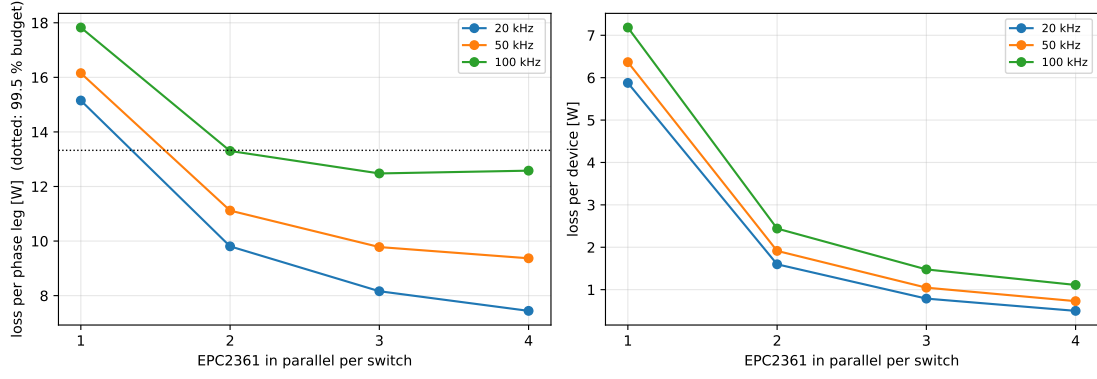


Figure 5: Loss per leg (left) and per device (right) versus devices per switch. The switching energy is scaled from the $n = 2$ LTspice $E(i)$ fit.

6 Loss model and switching-frequency selection

6.1 Loss model

Adapting [2] to a GaN leg with n devices per switch, sinusoidal current and dead time t_{dt} :

$$P_{\text{cond}} = \frac{I_{\text{rms}}^2 R_{\text{DS(on)}}(T_j)}{n}, \quad P_{\text{sw}} = f_{\text{sw}} \frac{1}{\pi} \int_0^\pi [E_{\text{on}}(i) + E_{\text{off}}(i)] d\theta, \quad i = \hat{I} \sin \theta, \quad (2)$$

$$P_{\text{dt}} = 2f_{\text{sw}} t_{\text{dt}} \overline{(V_{\text{SD}}(i) |i|)}, \quad P_{\text{Cu}} = \sum_k I_{k,\text{rms}}^2 R_k \quad (\text{Q3D DC matrix}), \quad (3)$$

plus the gate-drive and DC-link ESR losses. $E_{\text{on}}(i)$ and $E_{\text{off}}(i)$ are quadratic fits of LTspice DPT runs with the extracted parasitics and the selected gate resistors, so the layout is inside the loss model. The V_{SD} of the GaN reverse conduction (≈ 2 V) makes the dead-time term non-negligible. A negative off-voltage would increase it further [1, 4], which is one reason for the 0 V off-state used here.

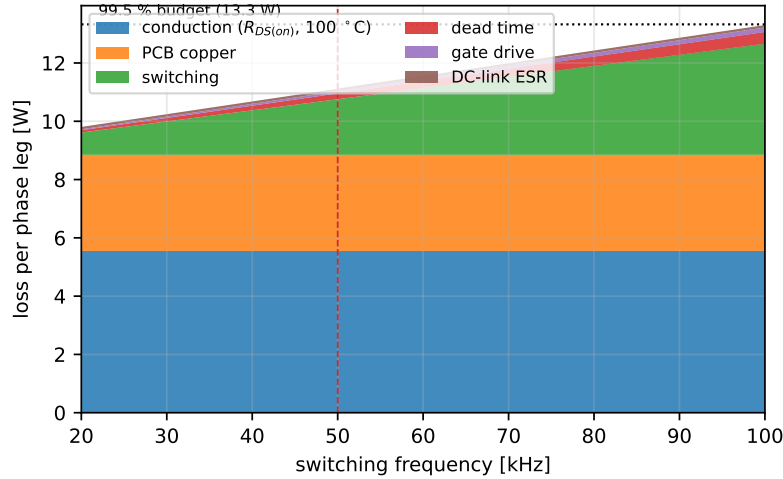


Figure 6: Loss breakdown per leg. Conduction and PCB copper are about 80 % of the loss at 50 kHz; switching is linear in f_{sw} and reaches the 99.5 % budget at 100 kHz.

6.2 Why a high switching frequency

Cooke and Rogers show a visible reduction of the phase-current ripple from 20 to 50 kHz with a GaN inverter [3], and Wattenberg et al. use the higher frequency to replace electrolytic capacitors with MLCCs [4]. Here the DC-link capacitance halves from 50 to 100 kHz (Section 10) at a cost of about 0.44 W per 10 kHz in switching loss.

Winding ripple requirement. The machine is not yet defined, so machine-side PWM losses are not evaluated here. The inverter side can, however, state what it requires: the peak-to-peak ripple of a phase is at most

$\Delta i_{pp} \approx V_{dc}/(4Lf_{sw})$, where L is the inductance seen by the ripple. For $\Delta i_{pp} \leq 10\%$ of the 141 A peak this means $L \geq 53, 27$ and $13 \mu\text{H}$ at 25, 50 and 100 kHz, a requirement to be handed to the machine design.

50 kHz is chosen as the nominal point; the block stays within the efficiency target up to 100 kHz, and the final value is fixed once the machine is selected.

7 Power-loop design

7.1 Analytical estimate and dominant parameters

For a vertical loop with forward and return currents on adjacent layers, the partial-inductance estimate used in [3, 1] is

$$L_{\text{loop}} \approx \mu_0 \frac{hl}{w}, \quad (4)$$

with dielectric thickness h , path length l and width w . For one cell ($l \approx 10$ mm, $w = 9$ mm) this gives 1.40 nH per mm of dielectric, i.e. 0.14 nH at $h = 0.1$ mm. Q3D gives a slope of 1.60 nH/mm, close to the analytical value. Q3D also shows a fixed part of 0.18 nH (vias, device pads and package) that no dielectric reduction can remove (Fig. 7). Table 5 lists the geometric parameters, following the parameter table of [1].

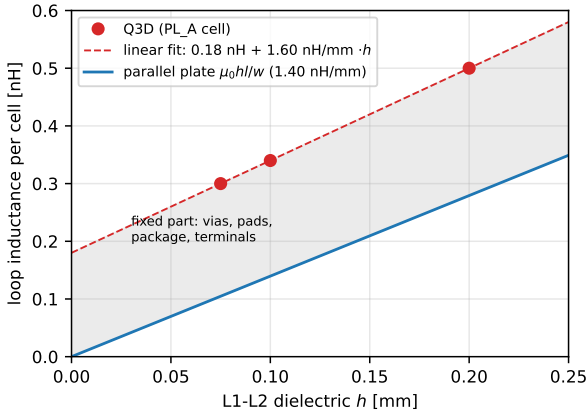


Table 5: Geometric parameters, constraints and choices.

Parameter	Effect	Constraint	Choice
h (L1–L2)	$L \propto h$	2 plies, ≥ 0.075 mm	0.1 mm
l (cap→QL S)	$L \propto l$	0805 + 2 FET pads	≈ 10 mm
w (cell width)	$L \propto 1/w$	board width	9 mm
via pitch	fixed part	drill rules	0.9 mm
local caps	ESL/ N	cell length	6×0805
copper L2–L4	R_{Cu}	cost, etching	2 oz

Figure 7: Loop inductance per cell versus the L1–L2 dielectric: analytical slope versus Q3D.

7.2 Layout and extraction

The block uses mirrored cells at $x = \pm 7.5$ mm. Each cell has its own vertical loop: the high-side drain row and six 0805 capacitors on L1, and the return on the DC– plane on L2, 0.1 mm below. The bulk MLCC rows above the cells feed both cells symmetrically. This follows the distributed-bank recommendation of [6], which avoids the cascaded, shared loop inductance that caused the 2 MHz resonance of the commercial module.

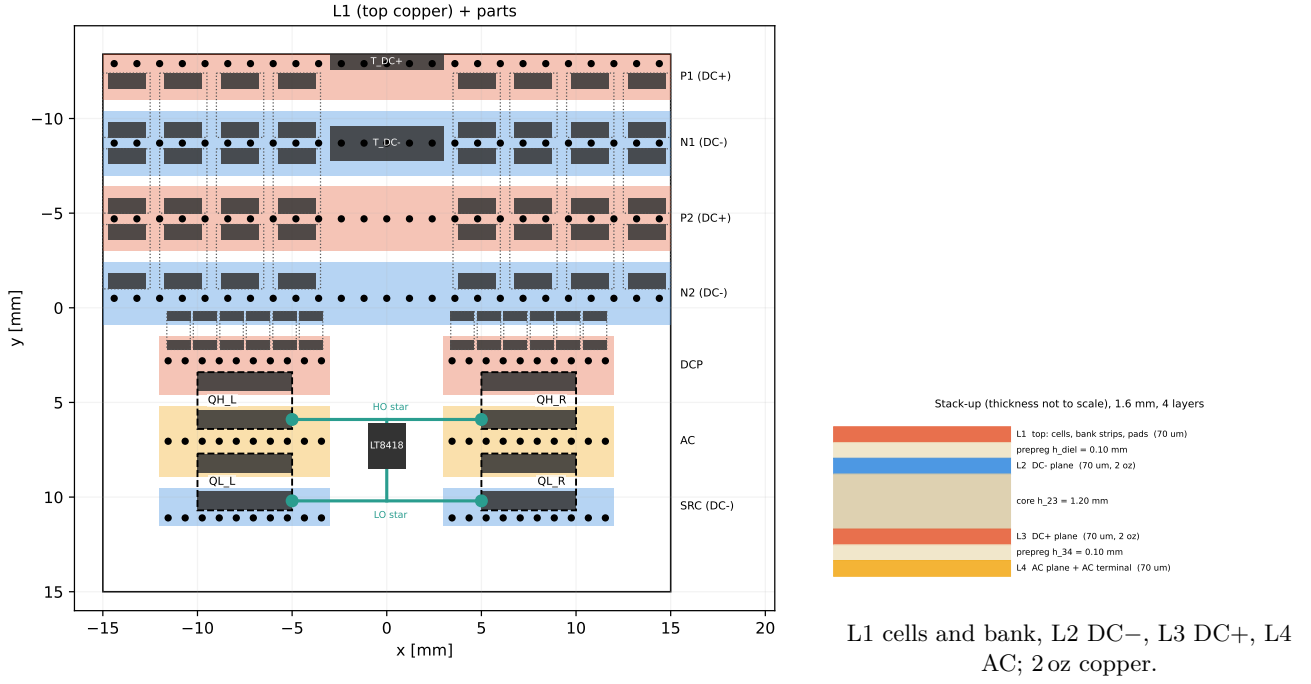


Figure 8: Layout generated from the same geometry as the Q3D model.

Q3D extracts the partial R/L matrix of the copper with every component as a gap between source and sink terminals. As in [1], the loop inductance follows from the matrix. For a branch-wise description, $L_{\text{loop}} = L_D + L_S + L_{ac} + 2(M_{DS} - M_{D,ac} - M_{S,ac})$. Here the full 21-terminal matrix is instead exported to LTspice as coupled inductors (K elements), and the effective loop is obtained by a nodal solve with both cells excited. The result is 0.312 nH copper and 0.374 nH with MLCC ESL per cell. Each FET sees this cell loop; paralleling halves the current and di/dt per loop, not the inductance. The EPC2361 package inductance and post-routing details are not included, so the measured loop will be somewhat higher.

7.3 Overshoot design space

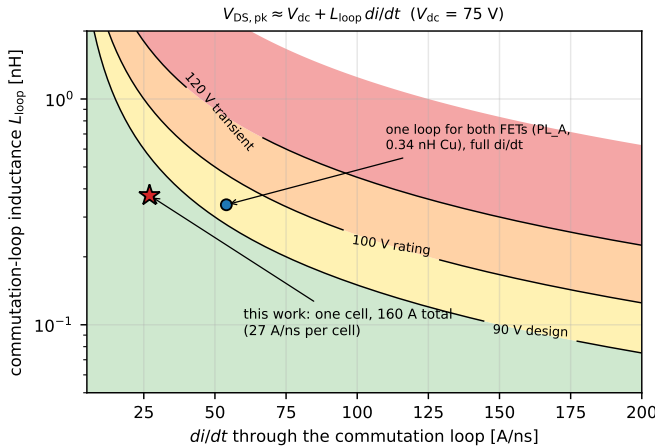


Figure 9: First-order overshoot map $V_{DS,pk} = V_{dc} + L_{\text{loop}} di/dt$. The operating point uses the effective di/dt through one cell (27 A/ns) that reproduces the LTspice peak at 160 A. At this di/dt the 90 V design limit allows $L_{\text{loop}} \leq 0.56$ nH per cell, which leaves room for the package and routing additions. A single loop carrying both FETs (PL_A) sees the full di/dt and would be at the 100 V rating.

8 Gate-drive design

8.1 Mechanisms

Tran et al. write the gate voltage of paralleled devices during di/dt and dv/dt as [1]

$$v_{gs,j} = v_{drv} - i_{g,j}(R_{g,j} + R_{s,j}) - (L_{g,j} + L_{sg,j}) \frac{di_{g,j}}{dt} - M_{gs,j} \frac{di_{D,j}}{dt} - L_{sg,j} \frac{d\Delta i_D}{dt}, \quad (5)$$

$$v_{gs,j} \approx v_{drv,off} + C_{gd} \frac{dv_{DS}}{dt} (R_g + R_s) + (L_g + L_{sg}) C_{gd} \frac{d^2 v_{DS}}{dt^2} \quad (\text{turn-off of the complementary switch}). \quad (6)$$

The first equation shows why mismatched quasi-common-source inductance (which drives a circulating current Δi_D) and power-to-gate coupling M_{gs} are the dominant risks for paralleled devices. The second gives the Miller bump. Fig. 10 maps every parasitic to its effect and to the layout rule adopted in this design.

parasitic	effect	rule in this design
power loop L_{loop}	V_{DS} overshoot, ringing, E_{off} ; limits I for 100 V dies [6]	vertical L1/L2 loop at 0.1 mm, local 0805s at QH, two short cells
common-source L_{CS}	negative feedback: slower di/dt , +47% E_{on} , gate bump	Kelvin return from each source pad; only package part remains
quasi-CSI mismatch ΔL_{QS}	circulating current Δi_D , gate ringing, 29% imbalance	mirror symmetry; equal source paths to the AC/DC- planes
gate loop L_G	V_{GS} overshoot if $R < 2\zeta\sqrt{L_G/C_{iss}}$; slower drive	driver in the corridor; $L_G = 1.7$ nH; flux-cancelling gate/return pair
gate-loop mismatch ΔL_G	turn-on skew $\approx \Delta L_G/R$; 4:1 E_{on} split at 2 vs 8 nH	centre driver: equal stars (side driver: 2.0 vs 7.7 nH)
coupling M_{gs} power→gate	induced V_{GS} during di/dt	gate traces on L1 inside the corridor, not over the switch node
C_{oss} , C_{PCB}	current spike, Q_{oss} loss $\propto n$	$n = 2$; small switch-node area on L4
shared DC-bank inductance	low-frequency resonance between cells and phases [6]	distributed bank feeding both cells; three legs on one module link

Figure 10: Parasitic–effect–rule map (after [5, 1]). Percentages are from the $2 \times$ EPC2361 sensitivity study (Section 9).

8.2 Gate-resistor design window

Treating the gate loop as a series RLC with C_{iss} gives a lower bound for the turn-on resistance (no V_{GS} overshoot) [1, 5, 7]. The Miller current through the turn-off path gives an upper bound for the turn-off resistance:

$$R_{G,on,tot} \geq 2\zeta \sqrt{\frac{L_G}{C_{iss}}} \quad (\zeta = 0.707 : 0.97 \Omega), \quad R_{G,off,tot} \leq \frac{V_{th}}{C_{rss} dv/dt} \quad (1.9\text{--}2.7 \Omega \text{ at } 32 \text{ V/ns}). \quad (7)$$

The selected totals are 3.0Ω on ($R_g + R_{G,int} + \text{driver pull-up}$, $\zeta = 2.2$) and 0.8Ω off ($R_{G,int}$ plus the shared pull-down, $\zeta = 0.58$). Both are inside the window (Fig. 11). Within the window, the turn-on value is set by the LTspice sweep (Fig. 12). Below 2Ω the high-side overshoot exceeds 90 V (112 V at 0.5Ω). Above it, E_{on} grows. $R_{g,off}$ hardly changes E_{off} because the turn-off is load-driven: the channel closes before V_{DS} rises and the load current charges C_{oss} ($dv/dt \approx I/2C_{oss,tot}$). $R_{g,off}$ therefore only weakens hold-off, and 0Ω is chosen.

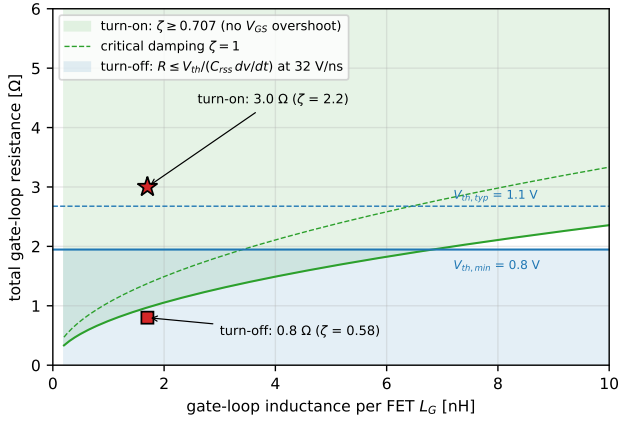


Figure 11: Gate-resistor design window for EPC2361 ($C_{iss} = 3.6$ nF, $C_{rss} = 13$ pF).

Table 6: Driver comparison (datasheet values).

Driver	Rating	Src/sink	Verdict
LMG1210	200 V	1.5/3 A	too weak
uP1966E	85 V	0.7/0.4 Ω	5 V margin
LT8418	100 V	4/8 A	baseline
2EDL5014AA	120 V	4/4 A	alternative

Requirements: bootstrap clamp (V_{SW} goes 2–3 V negative in dead time [4]), split outputs, CMTI ≥ 100 V/ns, one driver per switch (no driver-to-driver delay skew [1]).

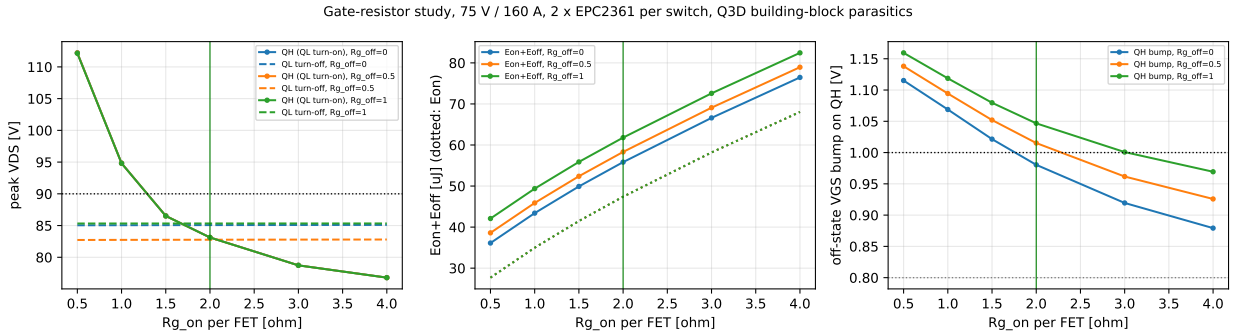


Figure 12: LTspice R_g study at 160 A with the Q3D parasitics. Selected: 2/0 Ω, giving QH 83.1 V, QL 85.1 V and a gate bump of 0.98 V.

Gate-bump margin (open risk). The simple Miller estimate gives 0.33 V. The LTspice bump is 0.98 V, which includes common-source and $L_G C_{gd} d^2v/dt^2$ terms. This is 89 % of $V_{th,typ}$ and above $V_{th,min}$ (0.8 V), so a low-threshold part could partially turn on for a few ns. The literature offers four mitigations:

- an external C_{GS} (270 pF in [4]), at some cost in speed;
- a small negative off-voltage (−1.5 V in [4], −3 V in [1, 3]), at the cost of dead-time loss and a bootstrap level shift;
- an active Miller clamp;
- a lower L_{CS} .

These should be evaluated in the next iteration and checked in the DPT.

9 Current sharing

Following [7, 8], a minimal two-FET LTspice model was perturbed one parameter at a time (Fig. 13). The ranking agrees with [1, 5]. Common-source mismatch without Kelvin returns and gate-loop mismatch dominate. V_{th} spread comes next. A drain-path mismatch is benign. The gate-loop case explains the centre driver. A mismatch delays one gate by about $\Delta L_G / R_{tot}$, roughly 2 ns, which is half the 4 ns current transition. On the building block itself, the two FETs carry 78.53 and 78.63 A at turn-off (Fig. 15).

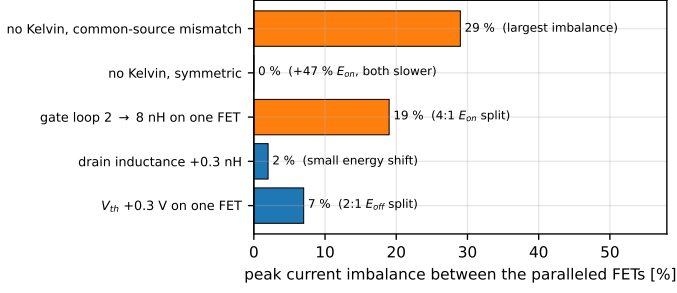


Figure 13: Sensitivity of the dynamic current imbalance (75 V, 120 A, $2 \times$ EPC2361).

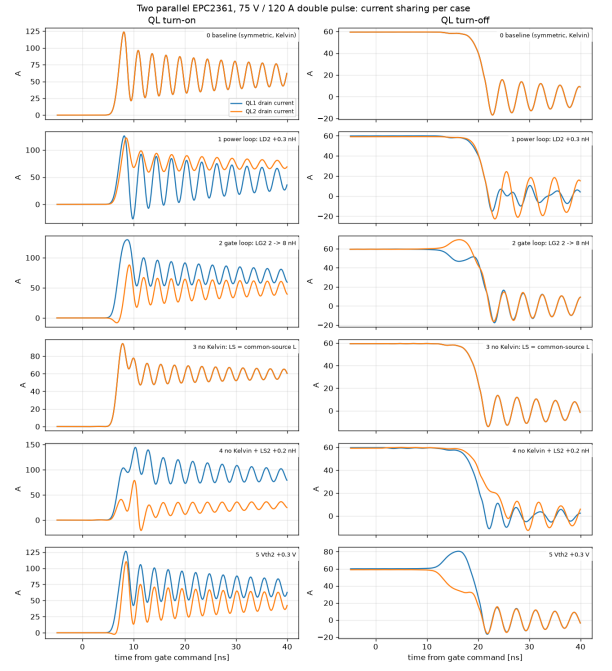


Figure 14: Waveforms of the six cases.

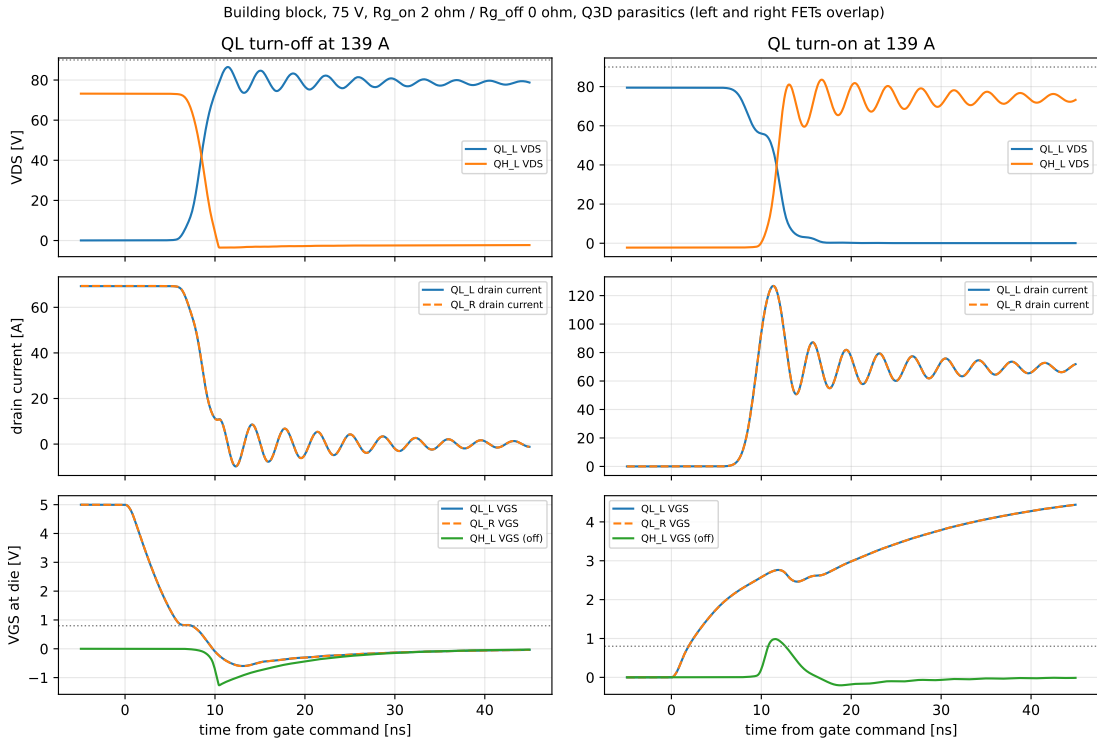


Figure 15: Building-block switching at the rated peak current. The left and right FET currents overlap.

10 DC-link design

As in [2], the capacitance is the larger of a voltage-ripple bound and a ripple-current bound, $C_{dc} = \max(C_v, C_i)$. With MLCCs, C_i is a count of parts rather than a capacitance: the HF ripple current (29 A_{rms} per leg at the rated point, up to 50 A) divided by the current rating of one capacitor (assumed 2.5 A, to be confirmed with the manufacturer's data). $C_v \approx \hat{I} / (4f_{sw}\Delta V_{pp})$ [4] is evaluated for three legs on a shared module link (78 μ F per leg at $\pm 2\%$, 50 kHz). Below about 98 kHz the bank is voltage-limited and shrinks as $1/f_{sw}$. Above that it is current-limited and stops shrinking (Fig. 16). This gives a natural upper bound on the useful f_{sw} for density. MLCC DC-bias derating is included (30 % of nominal at 75 V for the 1210 parts).

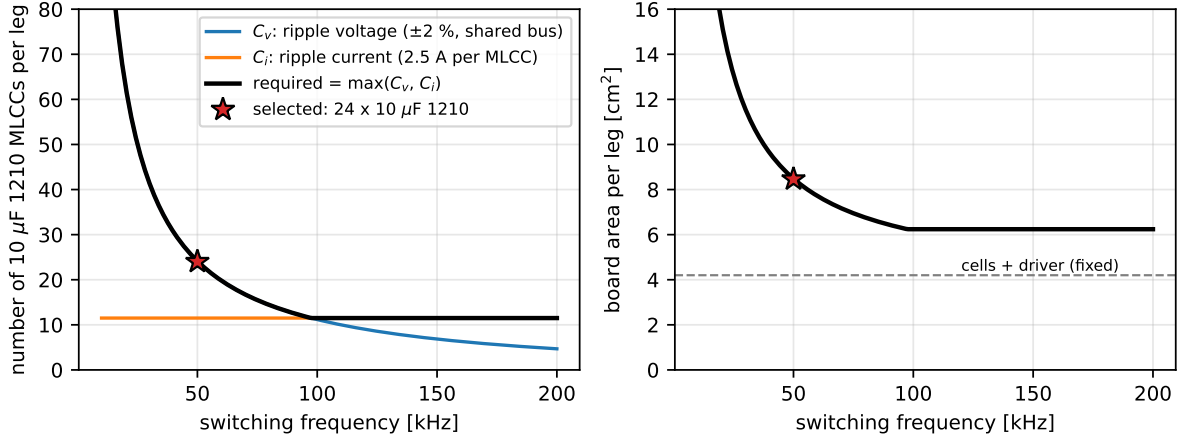


Figure 16: DC-link design space: MLCC count per leg (left) and resulting board area (right) versus f_{sw} .

DPT bulk capacitance. During the first pulse the bank must supply the load-inductor energy. Following [1]:

$$C_{DPT} \geq \frac{L_{DPT} I_L^2}{2V_{dc}\Delta V - \Delta V^2}. \quad (8)$$

For $L_{DPT} = 5 \mu\text{H}$ and 160 A ($t_1 = 10.7 \mu\text{s}$) this requires $575 \mu\text{F}$ for 2% droop and $233 \mu\text{F}$ for 5%. The on-board $77 \mu\text{F}$ is therefore not enough for the DPT, and an external low-inductance film or electrolytic bank of about 0.5–0.6 mF is needed.

11 Thermal design

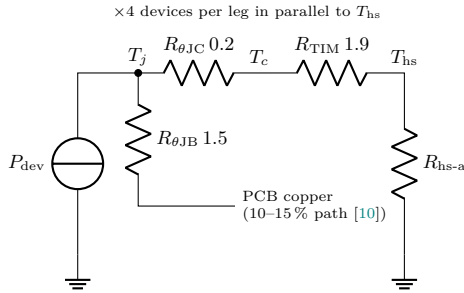


Figure 17: Thermal network per device.

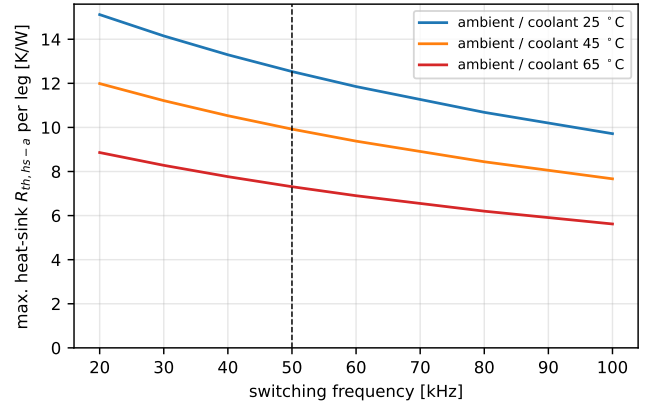


Figure 18: Maximum heat-sink resistance per leg for $T_j \leq 125^\circ\text{C}$.

Using the constraint of [2], $R_{hs} \leq (T_{j,max} - T_a - P_{dev}(R_{\theta JC} + R_{TIM}))/\sum P_{dev}$, the leg needs $R_{hs-a} \leq 7.3 \text{ K/W}$ with 65°C coolant at 50 kHz (9.9 K/W at 45°C). This is a mild requirement: with 0.5–1.9 W per device, a shared cold plate or a small forced-air heat sink suffices. The TIM, not the heat sink, dominates the per-device rise. For comparison, Cooke and Rogers measured a mean 16.9 K/W junction-to-ambient per HEMT on a 0.9 K/W heat sink with 8 devices per switch [3]. The PCB copper loss (3.30 W) is dissipated mainly in the board and must be included in the board temperature budget.

12 Efficiency–density–frequency co-optimisation

Zou et al. map efficiency and power density over (f_{sw}, P_o) and select the knee of the Pareto frontier [2]. The same is done here at board level, with the area model of Section 10 and the loss model of Section 6 (Fig. 19).

- Raising f_{sw} trades efficiency for density until about 98 kHz (0.42 kW/cm², 99.50 % at 100 kHz). Beyond that the DC link is current-limited, so density no longer improves and only efficiency falls. The frontier therefore ends at about 100 kHz.
- 50 kHz keeps 0.08 % efficiency margin at 0.31 kW/cm²; machine-side losses, which favour higher f_{sw} , are not included because the machine is not yet defined.

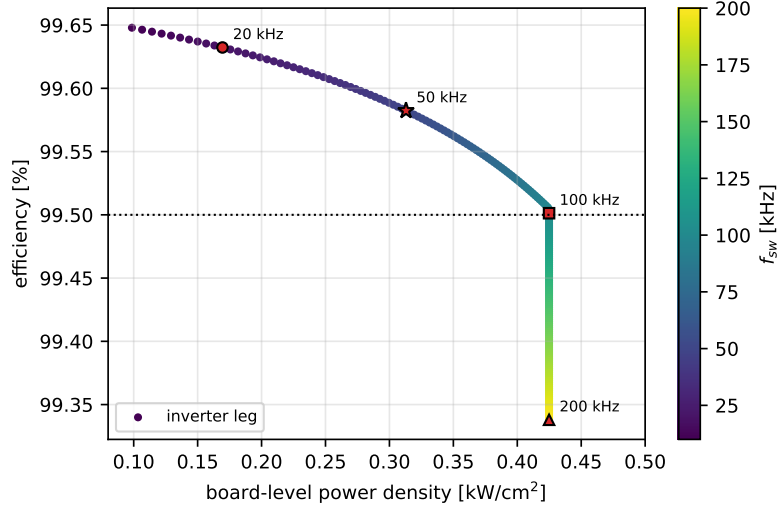


Figure 19: Efficiency versus board-level power density with f_{sw} as the parameter (10–200 kHz), inverter leg only.

13 Design summary and margins

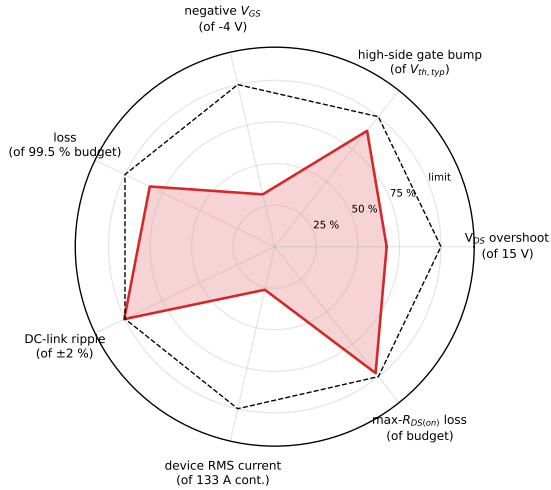


Figure 20: Utilisation of each constraint (1 = limit).

The radar (Fig. 20) shows where the design is tight:

- DC-link ripple at 100 % of the $\pm 2\%$ target (shared three-leg bus);
- loss with maximum $R_{DS(on)}$ at 97 % of the budget;
- high-side gate bump at 89 % of $V_{th,typ}$.

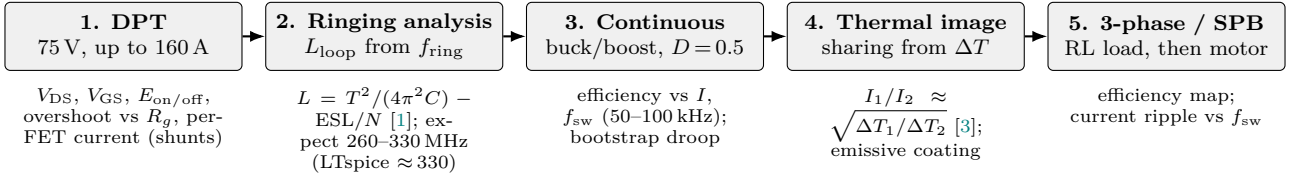
The voltage overshoot (67 % of the 15 V allowance), the negative gate excursion and the device current have comfortable margins. Comfortable margins are where the next iteration can trade for speed (lower $R_{g,on}$) or density.

Table 7: Design decisions.

Decision	Choice	Reason / evidence
Module voltage	75 V	100 V GaN with a 15 V design overshoot allowance (Fig. 9)
Devices per switch	$2 \times \text{EPC2361}$	knee of the loss curve; meets 99.5 % (Fig. 5)
f_{sw}	50 kHz (100 kHz capable)	efficiency margin at 50 kHz; density knee near 100 kHz; final value after machine selection (Figs. 6, 19)
Power loop	vertical, 0.1 mm, caps at QH, 2 mirrored cells	0.37 nH per cell, each cell carrying half the current (Fig. 1)
Gate drive	centre driver, Kelvin returns, LT8418	equal 1.7 nH stars; sharing < 0.1 % (Fig. 13)
Gate resistors	2Ω on / 0Ω off	inside the design window; $\leq 90 \text{ V}$ at 160 A (Figs. 11, 12)
DC link	$24 \times 10 \mu\text{F} + 12 \times 1 \mu\text{F}$	C_v -limited at 50 kHz; $\pm 2\%$ on a shared bus (Fig. 16)
Copper	2 oz on all layers	PCB copper $4.7 \rightarrow 3.3 \text{ W}$

14 Experimental validation plan

The reviewed works validate in a common sequence, from double-pulse tests to continuous and finally motor operation [1, 3, 4, 10]. The same sequence is planned here (Fig. 21).

**Figure 21:** Validation sequence and what each step verifies.

Measurement requirements. The turn-off edge takes about 2.4 ns (32 V/ns), and the cell loop rings at 260–330 MHz (0.374 nH with the C_{oss} of the off-state FET, 1.0 nF at 50 V to ≈ 0.6 nF at 75 V; LTspice shows ≈ 330 MHz). Probes and scope therefore need ≥ 1 GHz bandwidth with MMCX or tip-and-barrel connections at the device pads [4]. High-side V_{GS} needs an optically isolated probe. As noted in [6], terminal measurements under-read the die V_{DS} , so the test points sit at the QL drain and source pads.

15 Conclusions and next iteration

The building block reaches the efficiency target with margin at 50 kHz, remains at the target at 100 kHz, and achieves a power-loop inductance below the reviewed designs, with equal loops and gate stars for the paralleled FETs. The model-based flow makes the design reproducible and lets each change be traced to its effect on losses, overshoot and sharing. Planned improvements, ranked by benefit:

1. Gate-bump mitigation (external C_{GS} or small negative bias; driver with Miller clamp), verified in the DPT.
2. Copper-loss reduction: DC busbar contacts next to each cell and heavier L3/L4 copper (about -2 W). This makes 100 kHz possible at the efficiency of today’s 50 kHz design.
3. A larger or film DC link if the shared-bus ripple must be held at $\pm 2\%$ across all operating points, and a separate 0.5–0.6 mF bank for the DPT.
4. When the machine is selected, evaluate machine-side PWM losses in a separate study and confirm the switching frequency against the winding-ripple requirement.

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